MILITARY
SPECIFICATION

MIL-C-14255A(SigC)
29 March 1960
Superseding
MIL-C-14255(SigC)
23 February 1956

CENTRAL OFFICE, TELEPHONE, MANUAL AN/TTC-7( )
(UNITS OF)

1. SCOPE

1.1 This specification covers units of one type of central office that is a transportable manual telephone exchange designated as Manual Telephone Central Office AN/TTC-7( ). The central office consists of 3 switchboards, 3 line and trunk relay equipment cases, 3 main distributing frames, and necessary cables and accessories. It is equipped to provide 100 common battery lines, 100 common battery-local battery lines, 20 civilian office trunks. (See 6.3)

1.2 This specification covers the following units of the central office.

a. Switchboard, Telephone, Manual SB-249( )/TTC.

b. Telephone Circuit, Line Relay TA-223( )/TTC.

c. Telephone Circuit, Line Relay TA-226( )/TTC.

d. Telephone Main Distrubing Frame TA-257( )/TTC.

e. Cases

(1) Case, Chair CY-1490( )/T.

(2) Case, Cable Assembly CY-1498( )/T.

(3) Case, Accessories CY-1499( )/T.

(4) Case, Accessories CY-1500( )/T.

f. Cable Covers

(1) Cable Cover CW-323( )/T.

(2) Cable Cover CW-324( )/T.

g. Base, Stand MT-1132( )/T.

h. Night Alarm Bell.

FSC 5805

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i. Cable Assemblies

(1) Cable Assembly, Telephone CX-2584( )/U.

(2) Cable Assembly, Power, Electrical CX-3692( )/U and CX-3694( )/U.

(3) Cable Assembly, Power Electrical CX-4762( )/U (20-ringing).

j. Cords.

(1) Cord CO-38( ).

(2) Cord CO-258( ).

(3) Cord CO-393( ).

(4) Cord CO-409( ).

(5) Cord CO-413( ).

(6) Cord CO-414( ).

(7) Cord CX-4845( )/U.

k. Telephone jack covers.

l. Power Distribution Panel SB-1032( )/TTC.

m. Test Set, Telephone TS-1361( )/G.

n. Lead, Test CX-6100( )/G (FT-In)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.

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MILITARY (Cont'd)

MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For Use in Electronic, Communication, and Associated Electrical Equipment.)
MIL-P-11268	Parts, Materials and Processes.
MIL-C-12377	Central Office, Telephone, Manual AN/TTC-7( ), Packaging of.
MIL-M-13231	Marking of Electronic Items.
MIL-P-13905	Panel, Telephone Switchboard (Panel BD-132( )).
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SC-DL-142280	Line Relay Telephone Circuit TA-223( )/TTC.
SC-DL-142422	Stand Base, MT-1132( )/TT.
SC-DL-142424	Trunk Relay Telephone Circuit TA-226( )/TTC.
SC-DL-142425	Accessories Case CY-1499( )/T.
SC-DL-142455	Accessories Case CY-1500( )/T.
SC-DL-142457	Telephone Cable Assembly CX-2584( )/U.
SC-DL-142474	Telephone Main Distributing Frame TA-257( )/TTC.
SC-DL-142503	Chair Case CY-1490( )/T
SC-DL-142504	Cable Assembly Case CY-1498( )/T.
SC-DL-142699	Switchboard, Telephone, Manual SB-249( )/TTC.
SM-D-183295	Cable Cover, CW-323( )/T.
SM-D-183296	Cable Cover, CW-324( )/T.
SM-C-183439	Night Alarm Bell,
RL-B-312	Cord CO-38( ).
SC-D-4271	Cord CO-258( ).
SC-D-4309	Cord CD-413( ) and Cord CD-414( ).
SC-D-4310	Cord CD-393( )
SC-D-4421	Cord CD-409( )

(Copies of specifications and standards required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of ivitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification
Consolidated Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Description.- Manual Telephone Central Office AN/TTC-7( ) is a complete transportable telephone exchange consisting of 3 switchboard positions, and relay equipment for 200 lines, and 20 trunks with necessary power equipment provided for initial installation. The central office can be installed and put into operation rapidly, and can be dismantled and moved in a relatively short time.

3.2 Construction.-

a. Physical construction of the following units on contract shall conform to the procurement model.

(1) Electrical Power Cable Assembly CX-3692( )/U.

(2) Electrical Power Cable Assembly CX-3694( )/U.

(3) Electrical Power Cable Assembly CX-4762( )/U.

(4) Electrical Power Cable Assembly CX-4845( )/U.

(5) Power Distribution Panel SB-1032( )/TTC.

(6) Test Set, Telephone TS-1316( )/G.

(7) Lead, Test CX-6100( )/G (Ft-In).

(8) Telephone Jack Cover.

b. Construction of other units covered elsewhere herein shall be constructed in accordance with the drawings indicated.

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3.3 Procurement model.- A procurement model of the units listed in paragraph 3.2a will be available for inspection by prospective bidders and will be lent to the contractor.

3.4 Parts, materials, and processes; general.- In addition to any requirements of this specification covering parts, materials and processes for the units listed in 3.2a, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.4.1 Use of Virgin Materials in Plastic and Rubber Compounds for Insulation and Jacket of Wires and Cables.- Only virgin materials shall be used for plastic and rubber compounds intended for insulation and jacket of wires and cables.

3.5 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly or components.

3.6 Marking.-

3.6.1 General.- Marking shall conform to Specification MIL-M-13231.

3.6.2 Visibility.- Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers.- Units "l", "m", and "a" through "d" of paragraph 1.2 shall be serial numbered.

3.7 Tropicalization.-

3.7.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

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(2) It does not incorporate any material listed in 3.7.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.7.2 Treatment of materials.-

3.7.2.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flamability of treated material is held to the practical minimum.

3.7.2.2 Toxicity.- Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.7.2.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.7.2.4 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.7.3 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.

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(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.8 Finish, protective.- Units shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency.

3.9 Switchboard, Telephone, Manual SB-249( )/TTC.- Switchboard, Telephone, Manual SB-249( )/TTC shall be constructed in accordance with Drawing SC-DL-142699.

3.9.1 Description.- Manual Telephone Switchboard SB-249( )/TTC is a transportable manual unit having 400 face appearances for CB or CB-LB lines, 180 face appearances with designation strips for CB or CB-LB Lines. The 180 face appearances with designation strip are also suitable for ringdown trunks or civilian office trunks. When properly connected, a four-panel multiple switchboard can accomodate 1000 CB and LB lines and 160 trunks.

3.9.2 Electrical requirements.-

3.9.2.1 Circuits.- Each Switchboard shall provide the following circuits in the quantities listed below:

400 circuits for CB or CB-LB lines.
180 circuits for CB-or CB-LB lines with designation strip (also suitable for use as central office trunks)
15 Cord circuits
1 Dial Circuit
1 Ringing circuit
1 Operator's telephone and position grouping circuit
1 Power cable circuit
1 conference circuit of 10 jacks with designation strip
1 conference circuit of 9 jacks and 1 recall jack with designation strip.

3.9.2.2 Relay operation.- Supervisory relays of the cord circuits shall operate over lines with loop resistances of 0 to 1500 ohms with a line leakage resistance from infinity to no less than 10,000 ohms.

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3.9.2.3 Transmission requirements.

a. Insertion loss.- No combination of voice-frequency transmission circuits of this equipment in a completed systems connections, such as a line-cord or line-cord trunk combination shall have an insertion loss of more than 1.5 db at 1000 cycles per second or an insertion loss of more than 2.5 db at any other frequency between 300 and 3500 cycles per second. The insertion loss of individual circuits shall conform to requirements specified in Table I. The loss shall be measured as specified in 4.5.4.1.

b. Bridging loss.- The bridging loss of the operator's telephone circuit in the monitoring condition shall be an absolute minimum and shall not exceed the requirements specified in Table I. The bridging loss shall be measured as specified in 4.5.4.2.

Table 1 - Transmission requirements

(1000 cycle loss between 600 ohm lines)

Trunk Circuit, Two Way to Dial or Manual CO, Relay CO operated	0.4 db
Operator's Telephone and Position Grouping Circuit	0.3 db
(GRP switch normal)	
MON-MR switch in MON position (monitoring condition)	0.3 db
MON-MR switch normal (talking conditon)	1.4 db
Universal Cord Circuit (all switches normal)	
Relays RS and FS normal (magneto)	0.2 db
Relays RS and FS operated (common battery)	0.9 db
Relays RS normal and FS operated	0.6 db
Relays RS operated and FS normal	0.6 db

c. Transmission loss of operator's telephone circuit.

(1) Receiving loss.- The receiving loss through the operator's circuit shall not exceed 6.0 db when measured as specified in 4.5.4.3.1.

(2) Transmitting loss.- The transmitting loss through the operator's circuit shall not exceed 6 db when measured as specified in 4.5.4.3.2.

3.9.2.4 Crosstalk and induction.- The cable forms and wiring shall be free of crosstalk and induction in excess of that specified in 3.9.2.4a. Wires carrying ringing current shall be as far removed as practicable from wires carrying talking current and shall be laced on the outside of handformed switchboard cabling.

a. Crosstalk.- Crosstalk loss between voice-frequency circuits shall be 60 db or more at all frequencies between 300 and 3500 cycles per second when measured as specified in 4.5.4.4.

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3.9.2.5 Noise level.- The noise level in any voice-frequency transmission circuit shall not exceed 18 db RN (db above reference noise) when measured as specified in 4.5.4.5.

3.9.2.6 Phase distortion.- To meet requirements for transmitting facsimile signals, the phase distortion coefficient in the frequency band between 600 and 3200 c.p.s. shall not exceed 200 microseconds for any "trunk-cord-trunk" connection. This is defined as the difference between the maximum and minimum envelope delay and shall be measured as specified in 4.5.4.6.

3.10 Line Relay Telephone Circuit TA-223( )/TTC.- Line Relay Telephone Circuit TA-223( )/TTC shall be constructed in accordance with Drawing SC-DL-142280. When using L.B. the equipment shall operate over lines with loop resistance from 0 to 3500 ohms with line leakage resistance from infinity to no less than 10,000 ohms.

3.11 Trunk Relay Telephone Circuit TA-226( )/TTC.- Trunk Relay Telephone Circuit TA-226( )/TTC shall be constructed in accordance with Drawing SC-DL-142424. This equipment shall operate over trunks with loop resistance from 0 to 1500 ohms with line leakage resistance from infinity to no less than 10,000 ohms.

3.12 Telephone Main Distributing Frame TA-257( )/TTC.- Telephone Main Distributing Frame TA-257( )/TTC shall be constructed in accordance with Drawing SC-DL-142474. The unit shall be free from shorts, crosses, grounds, and opens when tested in accordance with 4.5.3.

3.13 Cases.- The cases listed below shall be constructed in accordance with applicable drawings and shall accomodate designated equipment and accessories.

3.13.1 Chair Case CY-1490( )/T.- Chair Case CY-1490( )/T shall be constructed in accordance with SC-DL-142503.

3.13.2 Cable Assembly Case CY-1498( )/T.- Cable Assembly Case CY-1498( )/T shall be constructed in accordance with SC-DL-142504.

3.13.3 Accessories Case CY-1499( )/T.- Accessories Case CY-1499( )/T shall be constructed in accordance with SC-DL-142425.

3.13.4 Accessories Case CY-1500( )/T.- Accessories Case CY-1500( )/T shall be constructed in accordance with SC-DL-142455.

3.14 Cable Assemblies.- The cable assemblies listed in paragraph 1.2(i) shall meet the requirements of Specification MIL-C-3885.

3.14.1 Telephone Cable Assembly CX-2584( )/U.- Telephone Cable Assembly CX-2584( )/U shall be constructed in accordance with SC-DL-142457.

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3.15 Cords.- The cords listed below shall be constructed in accordance with the respective drawing.

Cord CO-38	RL-B-312
Cord CO-258-A	SC-D-4271
Cord CD-393	SC-D-4310 Group 2
Cord CD-409	SC-D-4421
Cord CD-413	SC-D-4309
Cord CD-414	SC-D-4309

3.16 Cable covers.- Cable covers CW-323( )/T and CW-324( )/T shall be constructed in accordance with SM-D-183295 and SM-D-183296 respectively.

3.17 Stand Base.- Stand Base MT-1132( )/TT shall be constructed in accordance with SC-DL-142422.

3.18 Night Alarm Bell.- Night Alarm Bell shall be constructed in accordance with SM-C-183439.

3.19 Power Distribution Panel SB-1032( )/TTC.- Power Distribution Panel SB-1032( )/TTC shall meet the power requirements and performance of MIL-P-13905 except the alarm and 24 volt facilities are hor Part of the SB-1032( )/TTC.

3.20 Test Set, Telephone TS-1361( )/G.- Test Set, Telephone TS-1361( )/G shall have a loop circuit resistance of 1000 ohms and a shunt circuit resistance of 100 ohms when tested in accordance with 4.5.4.7.

3.21 Lead, Test CX-6100( )/G.- Lead, Test CX-6100( )/G shall meet the requirements of Specification MIL-C-3885.

3.22 Insulation resistance.- All circuits, when subjected to the test 4.5.2 shall have an insulation resistance of not less than 100 megohms.

3.23 Power Source.- All circuits shall operate on 44 to 54 volts supplied from a storage battery of twenty four lead-acid cells.

3.24 Service conditions.- The equipment shall meet the following service conditions:

3.24.1 Operation.- Continuous use for a period of 1 year, under the worldwide environmental conditions specified in 3.24.2 through 3.24.6 with no more than normal maintenance and replacement of parts. Where the contractor has to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.27) shall be considered as compliance with this requirement.

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3.24.2 Temperature (see 4.6.1).-

(a) Operating: Ambient temperature in the range of +125°F to +32°F. (The 125°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.24.3 Relative humidity (see 4.6.2).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.24.4 Elevation (see 4.6.6).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.24.5 Orientation (see 4.6.5).-

(a) Operating: Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left, or right).

(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.27) shall be considered as compliance with this requirement.

3.24.6 Vibration and bounce.- As described in 3.25.

3.25 Vibration and bounce.-

3.25.1 Vibration, internal.- With the equipment shock-mounts blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.6.4)

3.25.2 Bounce.- The equipment shall be capable of meeting the requirements of table II.

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Table II Bounce

Paragraph heading	Par. No.	Performance after test
Bounce;	4.6.3	Specified performance. No physical damage except minor surface abrasion.

3.26 Preconditioning.- Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.27 Preproduction samples.- The contractor shall furnish preproduction samples for approval if and as required by the invitation for bids and contract. (See 6.2(e)(1)).

3.28 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.2(i))

3.29 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.7.2) Individual items shall not be hand-picked for fit or performance. Matched pairs or sets, when permitted may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc.

3.30 Workmanship.- The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
  Slack.
  Protection.
  Clearance.
  Splicing.
  Connections, general
  Grounding, general
  Shielding on wire and cable.
Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Mechanical operation of controls.
Gears.
Self-tapping screws.

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(b) In this specification:

Tropicalization

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility.- Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.8.)

(1) Procurement inspection of items before preparation for delivery. (See 4.3)

(2) Procurement inspection of preparation for delivery. (See 4.8 and section 5)

4.2 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III and the group A and group B inspection specified in table IV and V respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) humidity, (4) temperature; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table III - Preproduction Inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Temperature	3.24.2	4.6.1
Bounce	3.25.2	4.6.3
Vibration	3.25.1	4.6.4

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4.3 Procurement inspection of items before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.3.1 through 4.3.2.1. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified herein, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.7.) Each unit which will be subjected to group A, or group B inspection shall be preconditioned after final assembly.

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.5.3) shall be last.

Table IV - Group A Inspection

Inspection	Reqt. Par.	Insp. Para.	AQL Major	AQL Minor
			(each item separately)	
Visual and mechanical			1.0%	4.0%
Cases, cable covers & stand base	3.13,3.16&3.17	4.7	2.5dphu	10.0dphu
Swbd, Tel, Manual SB-249( )/TTC	3.9	4.7	1.5dphu	6.5dphu
Tel, Cct, Line Relay TA-223( )/TTC	3.10	4.7	1.5dphu	6.5dphu
Tel.Cct, Trunk relay TA-226( )/TTC	3.11	4.7	1.5dphu	6.5dphu
Test Set, Telephone TS-1361( )/G	3.20	4.7	1.5dphu	6.5dphu
Tel. main dist. Frame TA-257( )/TTC	3.12	4.7	1.0%	4.0%
Cable assemblies	3.14	4.7		
Electrical				
Swbd, Tel., Manual SB-249( )/TTC				
Wiring and adjustment	3.9.2.1,3.9.2.2	4.5.1	4.0	
Insulation resistance	3.22	4.5.2	dphu	
Tel. Cct, Line relay TA-223( )/TTC				
Wiring and adjustment	3.10	4.5.1	2.5	
Insulation resistance	3.22	4.5.2	dphu	
Tel. Cct. Trunk relay TA-226( )/TTC				
Wiring and adjustment	3.11	4.5.1	2.5	
Insulation resistance	3.22	4.5.2	dphu	
Tel. main dist. Frame TA-257( )/TTC				
Wiring	3.12	4.5.1	2.5	
Insulation resistance	3.22	4.5.2	dphu	
Cable assemblies				
Wiring	3.14	4.5.1	1.5	
Insulation resistance	3.22	4.5.2	dphu	

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Inspection	Reqt. Par.	Insp. Par.	AQL Major	AQL Minor
AN/TTC-7( ) (Equipment set up)				
Insertion loss	3.9.2.3a	4.5.4.1	6.5 dphu for the group combined	
Bridging loss	3.9.2.3b	4.5.4.2		
Receiving loss	3.9.2.3c(1)	4.5.4.3.1		
Transmitting loss	3.9.2.3c(2)	4.5.4.3.2		
Crosstalk and induction	3.9.2.4	4.5.4.4		
Noise level	3.9.2.5	4.5.4.5		
Operational				
CB-LB Circuits	3.10	4.5.3.1	1.0%	
Civilian office trunk circuits	3.9.2.1	4.5.3.2		
Ringing circuits	3.9.2.1	4.5.3.3		
Busy signal	3.9.2.1	4.5.3.4		
Night Alarm	3.18	4.5.3.5		
Grouping key	3.9.2.1	4.5.3.6		
Conference circuit	3.9.2.1	4.5.3.7		
Test Set, Telephone TS-1361( )/G	3.20	4.5.4.7		

4.3.2 Group B inspection.- This inspection (including sampling) shall conform to Table V and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection.

4.3.2.1 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

Table V - Group B inspection

Inspection	Reqt Para.	Inspt. Para.
Interchangeability	3.29	4.7.2

4.4 Bounce preconditioning.- The components, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The components shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

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4.5 Electrical tests.-

4.5.1 Wiring and adjustments.- Circuits shall be tested for continuity, errors in wiring, conformity with wiring diagrams and operative readiness for service of all equipment. Tests shall be conducted in such a manner as to insure that all normal electrical and mechanical adjustments for operative service have been made on the components. These tests may be made with direct or alternating current; however, alternating current only shall be used in testing circuits normally operated in service by alternating current. The use of suitable automatic testing apparatus will be satisfactory.

4.5.2 Insulation resistance.- Insulation resistance between components and mountings of equipment circuit plates, between conductors or circuits, and between circuit conductors and equipment frames shall be tested to determine compliance with 3.22. A potential of 500 volts d.c. from a megger, or other approved instrument, shall be applied between test points. Before 500 volt potential is applied protector blocks, heat coils, fuses, and connections to ground punchings shall be removed if existent in unit under test. Whenever applicable the test shall be made from and through the answering jack on the line and trunk circuits in the switchboard unit.

4.5.3 Operational tests.- Manual Telephone Central Office AN/TTC-7( ) shall be setup with all equipment including relay cases, main distributing frames, telephone switchboards, and power panels properly cabled into the system. Tests outlined below require at least one man to perform telephone user operations at the main distributing frame and one man to act as operator at the switchboard positions. Switchboard operator should observe whether all incoming line lamps light when they should and if cord circuits function properly.

4.5.3.1 CB-LB circuits.-

(a) Set CB-LB toggle switches on Line Relay Telephone Circuit TA-223( )/TTC to the CB position. At the main distributing frame, cabled to Line Relay Telephone Circuit TA-223( )/TTC, connect two common-battery telephones on adjacent lines.

(1) When two telephones are interconnected check for voice transmission between telephone users and between each telephone user and operator; check ring-forward and ring-back operations and cord circuit supervisory lamps responding to hookswitch flashing.

(2) When above operations have been completed repeat the same procedure for two other common-battery line circuits and another cord circuit. Continue this procedure until all common-battery line circuits and all cord circuits have been checked.

(b) Set CB-LB toggle switches on Line Relay Telephone Circuit TA-223( )/TTC to the LB position.

(1) At the Main Distributing Frame (cabled to Line Relay Telephone Circuit TA-223( )/TTC, connect two magneto (local-battery) type telephones on adjacent lines. Crank the generator of one of the telephones and ask the operator for the line which has the other telephone connected to it. Operator may be informed of other line location by cranking generator of other telephone. When the two telephones

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are interconnected, check for voice transmission between telephone users and between each telephone user and operator; check ring-forward and ring-back operations and cord circuit magneto supervisory lamp operating in response to ring-off by each subscriber.

(2) When above operations have been completed, repeat the same procedure for two other local-battery circuits and another cord circuit. Continue this procedure until all local-battery circuits and all cord circuits have been checked.

4.5.3.2 Civilian office trunk circuits.-

(a) If a civilian telephone extension of a dial central office is available, connect one of the trunk circuits at the appropriate main distributing frame into the dial central office. Plug a calling cord into jack of circuit being tested and operate dial key. Dial tone should be heard. Operation of a dial wipe-out key should cause dial tone to be removed until this key is restored to normal. Restore dial wipe-out key to normal and complete a call by dialing a number. Check voice transmission from operator to called number and ask telephone user to call back by dialing your number. Pull down cord and await incoming signal from telephone user who is calling. When incoming signal appears, insert answer cord in jack and operate talk key to converse. Repeat this process until all cord circuits and trunk circuits have been checked.

(b) If an extension from a dial central office is not available, connect a magneto type telephone at the appropriate main distributing frame and crank the generator. This should bring in the incoming signal on the switchboard. After the operator has answered the call and checked voice transmission, remove telephone and replace it with an ohmmeter or other continuity indicating device. D. C. closure should be placed or removed from the line by operator plugging or unplugging into jack. Operation of dial wipe-out key should remove closure. Operation of dial with dial key operated while call cord is inserted, should interrupt closure at dial pulse rate. After above items have been checked repeat procedure until all cord circuits and trunk circuits have been tested.

4.5.3.3 Ringing circuit.- Select a telephone line circuit at one of the main distributing frames. Connect to this line circuit a telephone bell from tip to switchboard ground (positive battery) and from ring to ground at the main distributing frame. Insert a calling cord into the associated jack at the switchboard and operate the ring key with the MON-MR key in the normal position. The telephone bell connected from ring to ground should ring. With the ring key operated and the MON-MR key operated to the lock position, the telephone bell connected from tip to ground should ring. Repeat above test at every switchboard position using telering, emergency d.c. 20 cycle ringing, and hand generator for ringing power.

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4.5.3.4 Busy signal.- During tests described in 4.5.3.1, 4.5.3.2 and 4.5.3.3 above, when plug is in jack of one switchboard position, perform busy test by placing tip of call cord on sleeve of busy jack at each of the remaining switchboard positions which have this jack appearance. With the talk key operated a click should be heard in the operator's headset indicating busy signal. Check busy test operation at all switchboard positions.

4.5.3.5 Night alarm.- During tests described in paragraphs 4.5.3.1 through 4.5.3.3 above check night alarm circuit action. Night alarm should function every time an incoming signal appears, when night alarm switch is in the "on" position.

4.5.3.6 Grouping key.- Connect a telephone to a line circuit (either CB or CB-LB) at one of the main distributing frames. Check grouping key operation by operating grouping key and plugging cord with associated talk key operated from an adjacent switchboard position into jack which has telephone connected to it at the main distributing frame. Operator should be able to converse with telephone. Repeat this test until grouping key operation and every switchboard position has been checked.

4.5.3.7 Conference circuit.- In order to check conference circuit, proceed as follows: At the main distributing frame connect three (3) telephones on adjacent line circuits. Signal with one telephone and ask operator to connect caller with the other 2 telephones in a conference circuit. Operator may be informed of switchboard appearance by signalling with the other 2 telephones. When the 3 telephones are connected into the conference jacks, check to see that all telephone users can converse with each other. Move one of the plugs in the conference circuit along the strip until all remaining jacks of the first conference circuit are tested. Repeat the same procedure for the second conference circuit.

4.5.4 Transmission measurements.- Voice-frequency measurements specified below shall normally be made only on individual circuits and at the standard test-tone frequency of 1000 c.p.s., except in testing the preproduction sample. Design analysis or tests shall be made on sufficient circuits and combinations of circuits in the preproduction sample and at sufficient frequencies to ascertain that overall transmission characteristics are satisfactory and reasonably uniform with respect to the requirements of 3.9.2.3 between 300 and 3500 c.p.s.

4.5.4.1 Insertion loss.- The insertion loss of voice-frequency transmission circuit shall be measured at 1000 ±25 c.p.s. by means of a decibel meter (or a transmission measuring set) connected to the other end of the circuit. The test equipment shall present an impedance of 600 ohms ±10 percent at voice frequencies to the circuit being measured. Suitable circuits (or test jigs) shall be included to simulate the operated circuit condition (i.e. the correct d.c. circuit condition) during the test period. The test-tone power applied to the circuit being measured shall be 0 dbm ±.5 db. The difference in db between the transmitted and received power shall be the insertion loss of the trunk. The insertion loss shall conform to 3.9.2.3a.

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4.5.4.2 Bridging loss.- The bridging loss of each operator's telephone circuit shall be measured in essentially the same manner as the insertion loss except that the test-tone source and the transmission measuring equipment (with correct power level and frequency impedances) shall be connected together and the operator's telephone circuit shall be bridged across the connection. The difference in db between the received power with the operator's telephone circuit bridged across the connection and without the operator's telephone circuit shall be the bridging loss.

4.5.4.3 Transmission loss of operator's telephone circuit.- Operator's telephone circuit shall be tested for receiving loss and transmitting loss as specified below. The same test equipment, power level, impedance and frequency may be used as for insertion loss. The MON-MR switch shall be normal.

4.5.4.3.1 Receiving loss.- The signal generator shall be connected to the tip and ring of the operator's telephone circuit, a 20-ohm resistor shall be subsituted for the transmitter, and the transmission measuring set (connected through an impedance-matching pad or network so as to present an impedance of 275 ohms at voice frequencies to the circuit being measured) shall be substituted for the receiver. (A vacuum-tube voltmeter connected across a 275-ohm resistor can be used instead of the transmission measuring set if desired). The difference in db between the transmitted and received power shall be the receiving loss of the circuit and shall not exceed the value specified in 3.9.2.3c(1).

4.5.4.3.2 Transmitting loss.- The signal generator (connected through an impedance-matching pad or network so as to present an impedance of 20 ohms to the circuit being measured) shall be substituted for the transmitter, a 275-ohm resistor shall be substituted for the receiver and the transmission measuring set shall be connected to the tip and ring of the operator's telephone circuit. (Vacuum-tube voltmeter connected across a 600-ohm resistor can be used instead of the transmission measuring set if desired.) The difference in db between the transmitted and received power shall be the transmitting loss of the circuit and shall not exceed the value specified in 3.9.2.3c(2).

4.5.4.4 Crosstalk.- Crosstalk shall be measured for all combinations of circuits which might develop crosstalk in excess of the value specified in 3.9.2.4. The same test equipment, power level, impedance, and frequency may be used as for insertion loss determination except that the transmission measuring set shall be Transmission Measuring Set TS-559( )/FT or equal, using flat weighting. If desired, the measurement may be made by the vacuum-tube voltmeter method, if it can be demonstrated that the results conform to the requirements of 3.9.2.4a. During the measurements, two 600-ohm ±10 percent, non-inductive resistors, one for each circuit, shall be connected across adjacent ends of the two circuits being tested. The test tone shall be applied to the open end of one circuit and the crosstalk shall be measured at the open end of the other circuit.

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4.5.4.5 Noise level.- The noise level in each voice-frequency transmission circuit shall be measured with Transmission Measuring Set TS-559( )/FT or equal, using FIA line weighting, and shall conform to 3.9.2.5.

4.5.4.6 Phase distortion.- The phase distortion shall bemeasured normally only in testing the preproduction sample to determine whether there is anything inherent in the design which might interfere with the transmission of facsimile signals through a "trunk-cord-trunk" combination circuit in this equipment. The measurements shall conform to 3.9.2.6.

4.5.4.7 Test Set, Telephone TS-1361( )/G.- Test Set, Telephone TS-1361( )/G shall be tested to meet the requirements of 3.20.

4.6 Service condition tests.-

4.6.1 Temperature.- The equipment shall be subjected to the temperature cycle shown in Standard MIL-STD-169. Repeat enough of 4.5.3 during steps 1, 4, 9 and 10 to determine operational condition of equipment. The performance of the equipment shall be comparable to performance prior to temperature cycle.

4.6.2 Humidity test.- The equipment shall be tested as follows:

(1) Dry at 130° ±5° F. for 24 hours.

(2) Subject to continuous cycling for a minimum of 5 of the 48-hour cycles shown on Standard MIL-STD-170, and measure the performance during the periods indicated on that standard.

(3) Do not remove from the test chamber for measurements.

(4) Start measurements not more than 5 minutes after power has been applied. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.6.2.1 Performance.- The measurements made during the test shall be as given below. If any equipment fails to meet the indicated performance, it will be considered as failing the humidity test.

4.6.2.2 Repeat enough of the tests of 4.5.3 to determine the operating condition of equipment. The performance of the equipment shall not have degraded from that obtained prior to the humidity test.

4.6.3 Bounce test.- The major components, in their transit cases shall each be placed on the table of the Package Tester as made by the L.A.B Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ±1 percent for a total of 3 hours. During each 1/2 hour period of the test, the transit case shall rest on a different face.

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4.6.4 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second to determine compliance with 3.25 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1 cycle per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corp., Cambridge, Massachusetts or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.6.5 Orientation test.- The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left and right) to an angle of 20° + 3° - 1°. During inclination in each plane, the equipment shall meet the requirements of 3.24.5.

4.6.6 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, it shall meet the test requirements of this specification.

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.6.6(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.6.6(a). The equipment shall meet specified performance for these measurements.

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4.7 Visual and mechanical inspection.- The cases, stand base, and cable covers shall be examined for the defects listed in table VI. The wired equipment shall be examined for all defects listed in Standard MIL-STD-252 except type 1 defects, Soldering.

4.7.1 Visual inspection of soldering.- The wired equipment shall be examined for the defects listed in Standard MIL-STD-252 under type 1 defects, Soldering.

4.7.2 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.29. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, and mountings which receive mating parts of assemblies and major components.

(e) Location of connectors, locking pins, fasteners, and mountings on assemblies.

(f) Size and form of special threads.

Table VI - Classification of Visual and Mechanical Defects for Cases, Stand Base, and Cable Covers

Classification	Defects
Major	Partitions or blocks in cases will not allow stowage of items. Carrying handles, hinges, or cover latches not located properly. Inoperative cover latches, hinges, or handles. Assembled stand base does not stand firmly on all four legs. Cable covers do not conform to material and design of models. Finish not in accordance with applicable specification. Interchangeability.
Minor	Cover reinforcements missing or improperly assembled. Scratches, cuts, abrasions, etc., with exposure of bare metal. Difficult operation of carrying handles and cover latches of cases. Difficult assembly of stand base.

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4.8 Procurement inspection of preparation for delivery.- This inspection shall conform to Specification MIL-C-12377.

5. PREPARATION FOR DELIVERY

5.1 Packing, packaging and marking.- Packing, packaging and marking shall be in accordance with Specification MIL-C-12377.

6. NOTES

6.1 Intended use.- Manual Telephone Central Office AN/TTC-7( ) is designed for use at any headquarters requiring telephone facilities.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, or level B.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item listed in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.27)

(2) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.28)

(i) Quantity of tools and running spare parts required. (See 3.28)

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6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/TTC-7W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Location of operational inspection.- It is desirable that the operational inspection (4.5.3) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to shipment.

6.5 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 Inspection.- Inspection is the examination or testing, or both, of equipment to determine compliance with applicable requirements. Sampling is an element of inspection.

6.6.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.6.2 Testing.- Testing consists of determinations of compliance using technical means.

6.7 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make minimum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A. The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL-STD-105.

(b) Type B. That inspection set forth in the Quality Assurance Provisions of this specification, or the contract, reduced in amount under the reduced inspection provisions of Standard MIL-STD-105.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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